

Power Management Techniques for Advanced SoCs

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Agenda

- Designs and power trends
- SoC application trends
- Power optimization technologies
- ARM Power Management Kit
- Power management cost
- EDA collaboration
- Summary

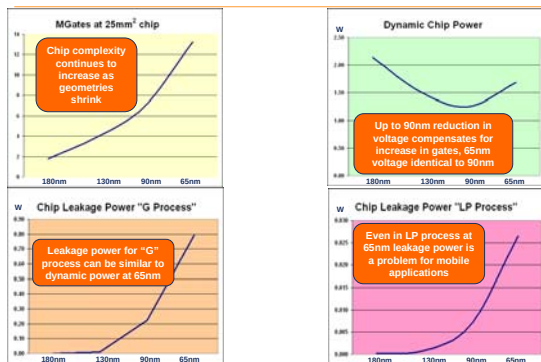


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Complexity and Power Explosion



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SoC Application Trends

Application	Application and Use Model	Requirements
Cell Phone Smart Phone 	- Talk - Take and send photos - Listen to music - Make and send video	- Extremely long standby time - Long battery life in usage
Laptop Handheld computer 	- Email, web browse - Office work - Gaming	- Very long battery life for email, browsing - Good performance for gaming
Portable Media Player 	- Read and store E-books - Listen to music - Watch video	- Very long battery life - Good video quality
GPS Unit 	- Display position - Calculate route	Long battery life in usage

Flexible use models, flexible performance requirements!

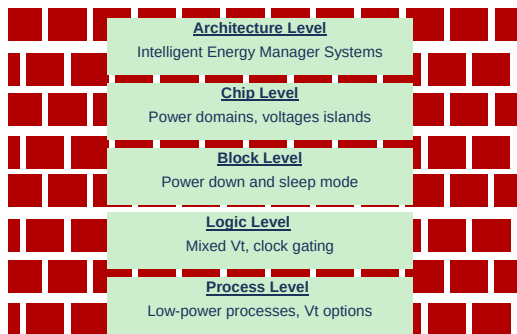


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Power Optimization Technology



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Physical IP: Low-Power Design

Flexible Use Models of Advanced SoC Designs



Flexible Power Optimization Technologies

- Sleep modes
- Retention
- Low-power modes
- Voltage scaling
- Back-biasing



Physical IP for Power Management



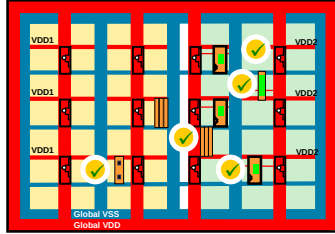
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ARM Power Management Kit (PMK)

- Power gates (MT-CMOS)*
 - Power control of voltage islands via switchable voltage rails using header (shown) or footer cells
- Level shifter and isolation cells
 - Up and down shifting with optional enable signal
- Retention flip-flops*
 - Maintain FF state after power down for leakage reduction
- Back-bias support*
 - Reduce leakage current via well-biasing with special fill_tie cells
- Always on buffer*
 - Buffering of signals in powered-down areas



*Note: requires tapless library

Note: Picture shows a conceptual implementation

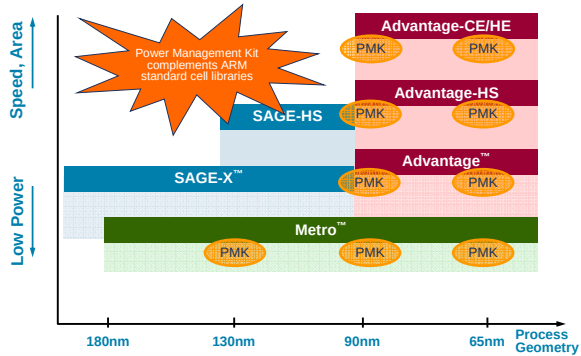


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Standard Cell Families and PMK

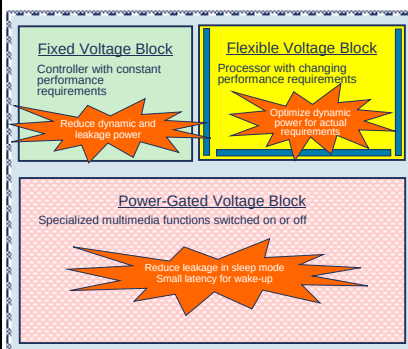


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SoC Power Optimization

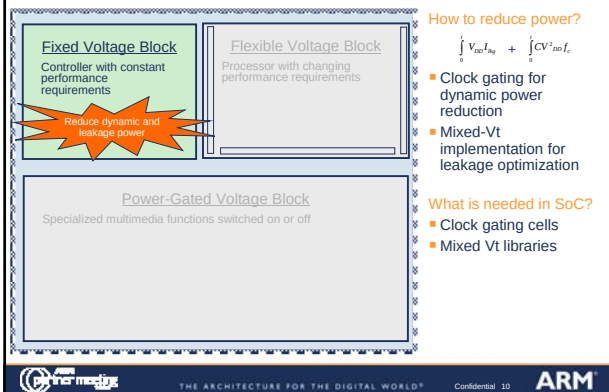


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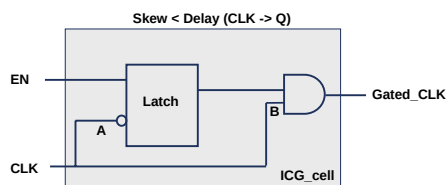


SoC Power Optimization

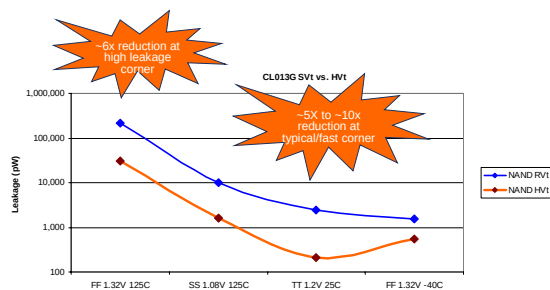


Integrated Clock Gating (ICG) Cell

- Clock registers only when needed
- Decreased switching activity reduces dynamic power
 - Typical chip level power savings of >30%
- ARM libraries have scan and non-scan version of the ICG cell (8-drive sizes each)
- Inserted automatically by DC / RTL compiler during synthesis



Benefits of Multi-Vt



SoC Power Optimization

Fixed Voltage Block
 Controller with constant performance requirements

Flexible Voltage Block
 Processor with changing performance requirements
 Optimize dynamic power for actual requirements

Power-Gated Voltage Block
 Specialized multimedia functions switched on or off

How to reduce power?

$$\int_0^T C V^2 \omega f_c dt$$

- Reducing voltage from 1.2V to 0.8V saves over 50% of dynamic power!

What is needed in SoC?

- Level shifters to ensure correct functionality with other blocks
- Low voltage characterization to ensure performance at different voltage levels

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Level Shifters

Island A Low Voltage Domain

Low Voltage Signal → Down shifter → High Voltage Signal

Island B High Voltage Domain

High Voltage Signal → Up shifter → Low Voltage Signal

VDDI

- Level Shifters provide "shift up", "shift down" functionality
 - Proper interface between islands of different voltage levels
 - Available with and without enable/clamp signal and with different drive sizes
- Dual-voltage characterization for all level shifters
 - Separate voltage value for input and output voltage

- Isolation cells
 - Used to isolate switchable power islands with identical voltage levels
 - Isolation cells are similar to HL level shifters with enable
- Fail-safe shifter and isolation cell design allows flexible power-on and power-off sequences

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Extensive Low Voltage Characterization

Example for 1.2V process:

Nominal Vdd	1.2V	1.0V	0.8V
Best Case P=FF, T = -40C	1.32V	1.1V	0.88
Best Case P=FF, T = 0C	1.32V	1.1V	0.88
Typical Case P=TT, T=25C	1.2V	1.0V	0.8
Worst Case P=SS, T=125C	1.08V	0.9V	0.72
Leakage Worst Case P=FF, T=125C	1.32V	1.1V	0.88

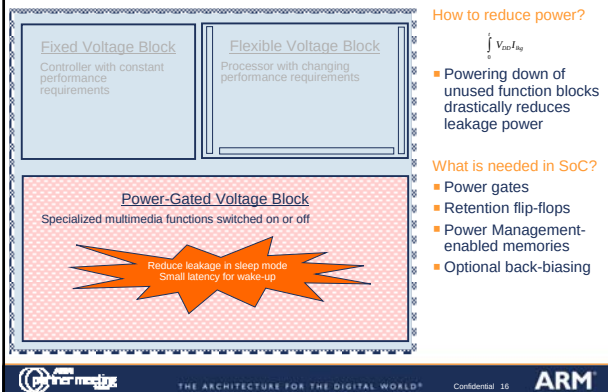
Standard Characterization Additional Characterization corners

Accurately derate any voltage between two characterization points

1.08V 0.9V 0.96V

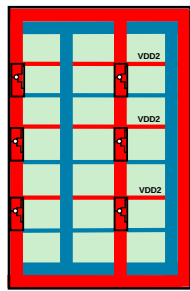
- CCS or ECSM models for accurate simulation over continuous voltage range

SoC Power Optimization



Power Gating

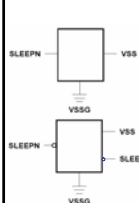
- Course grain power switches
 - Switching power or ground for multiple cells at block level
 - Disconnect local power from global power via power gates (MTCMOS switches)
 - Different implementations possible
 - Switches on the connection power strip and power rail
 - Switch power ring around block (similar to memory)
 - Reduced area overhead compared to other solutions
 - Analysis of the power network (to determine voltage drop) recommend
 - ECSM or CCS views can be used to calculate accurate timing (interpolating of voltage levels)



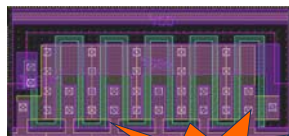
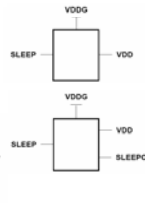
Switches on the connection power strip and power rail to switch VDD2

Power Gating

Footer Switches



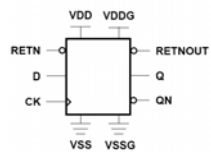
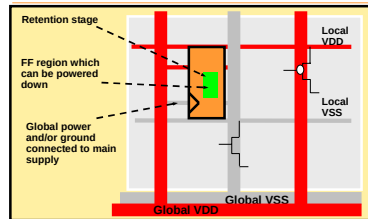
Header Switches



Example Footer Switch – same cell height as standard cells

- Header and Footer switches allow flexibility
 - Header to switch VDD, Footer to switch VSS
 - Same cell height as normal cells
 - Different transistor sizes (cell width) available
- All switches are implemented with or without buffered control signal

Retention Flip-Flop



Note: Picture shows a conceptual implementation

- Maintain FF state after power down
 - Extreme low leakage without loss of flip-flop state
- Retention flip-flops consists of main stage and storage stage
 - Main stage is connected to switchable (local) power rail, standard or low Vt
 - Storage stage (high Vt) is connected to global power/ground to maintain flip-flop state with minimal leakage current

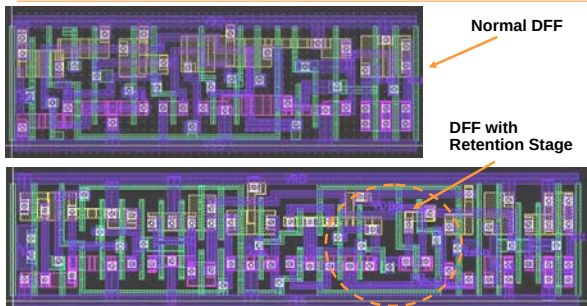


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Retention Flip-Flop Design



- Similar performance as normal FF
- Retention capability requires area trade-off
- Slightly higher dynamic power

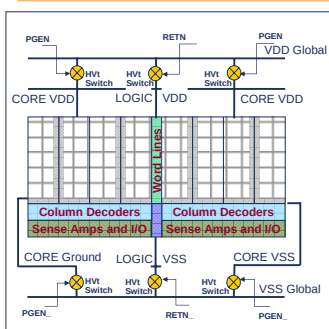


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Memory Power Management



- Special power modes through integrated power gating
 - Standby mode to control dynamic power
 - Retention mode to minimize leakage power
 - Shutdown mode
- Features
 - Separate core and logic power supply nets
 - Power gating HVT switches integrated in the power rings
 - Placed and optimized for performance, low power
- Leakage optimization for standard operation mode
 - Multi-Vt design
 - Dramatically reduces leakage with only a nominal drop in performance



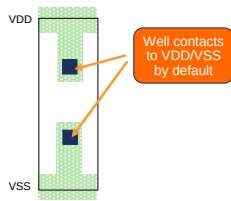
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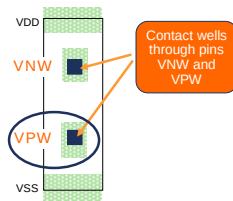
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Special Fill Cells for Back-Biasing

Standard FILL_TIE



FILL_TIE_BB



- Enables back or forward biasing for performance/leakage optimization
 - N-well voltage different from VDD
 - P-well voltage different from VSS (triple well process)
- Bias voltage routed as signal pin

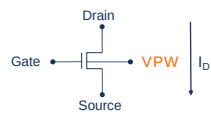
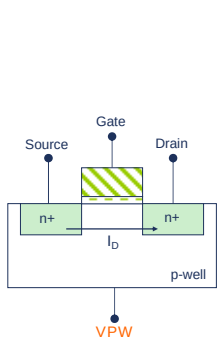


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NMOS Example with Well Access



- Control voltage on 4th terminal: **VPW**
 - Connect to NMOS Bulk through **VPW** pin in Special Fill Cells
 - When **VPW** < Source
 - V_{th} increases
 - I_D decreases
- Leakage **Decreases**



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Power Optimized Designs Enabled

	Dynamic Power Control	Leakage Power Control
Clock gating standard cells → no dynamic power on FF's	✓	
Extended operating range → Lower voltages for timing uncritical blocks → Voltage level shifters as interface → Digital voltage and frequency scaling (DVFS) support	✓	✓
Multi Vt support in standard cells and memories → use high Vt (low leakage) cells with reduced performance in uncritical paths		✓
Power gating and power down modes in memories → Flexible power saving modes		✓
Power gates for logic blocks → Shut-down of parts of the design		✓
Retention flip-flops → Minimal leakage current by maintaining FF state w/o loss of performance in normal mode		✓
Back-bias support → Minimal leakage current for inactive or low performance blocks		✓

Flexible combination of Power Management functions depending on design goals!



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Power Management and Cost

PMK Component	Advantages/Disadvantages
Clock-gating	+ No additional cost + Support by all EDA tools
Multi-Vt	+ Support by all EDA tools + No effect on critical path + Good leakage power reduction - Additional mask layer(s)
Level-shifters	+ Easy implementation + Support by major EDA tools - Moderate area increase
Power-gates	+ Significant leakage savings in inactive modes - 10 to 15% area increase - Careful control of power-on sequence required
Retention FF's	+ No performance loss in active mode + Fast energy efficient restore of data (alternative solution with scan-out/in has large latency, higher energy consumption) - 10 to 30% FF area increase
Back-Bias	+ Easy implementation for leakage savings in inactive mode - Requires additional voltage source(s) - Limited leakage savings compared to power-gating



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Low Power EDA Design Flow Support

- Close collaboration between ARM and EDA companies during development of tools and Physical IP
 - Regular technical meetings with EDA companies
 - Optimize IP and tools to ensure compatibility
 - PMK cell development and emerging tools ensure that cells can be used in an all phases of the design flow
 - Reference flow development
- Wide range of EDA view support
 - Accurate simulation of low voltage operating points
 - Accurate power estimation with support for standard and emerging EDA tools
 - Extensive model support ensures that all power design challenges can be addressed, e.g., state-dependent leakage power



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PMK Flow in Major EDA Tools

System Specification and Architecture	Designate islands and voltages
Write RTL / Synthesize	RTL constructs for power off / gate and data retention
Formal Verification	Verify retention flops and P/G connections
Verilog Sims	Shows "retain" and "isolate" behavior
Floorplan / Power Grids, Islands, Gates	Placement of power gates and dynamic IR drop analysis
Place and Route	Placement of PMK cells, routing of control signals
Formal Verification	Verify retention flops and P/G connections
Back-annotated Verilog Sims	P/G Connected netlist shows Xs when power gated
Timing Closure	Validate timing at multiple voltage conditions
DRC / LVS	Verify P/G connections and design database
Tapeout!	

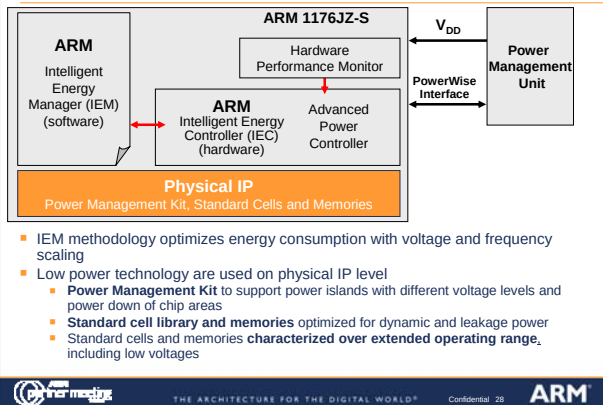


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ARM Intelligent Energy Management (IEM)



Comprehensive Power Management

- Support for broad range of sophisticated power management techniques
 - Active control of dynamic power
 - Active control of static / leakage power
- Power Management Kit available as add-on for Metro™ and Advantage™ ARM standard cell libraries
- Power management integrated in Metro and Advantage memories
- Close collaboration with EDA partners
 - Reference methodologies
 - Modeling and automation
- Used by the ARM Intelligent Energy Manager technology (IEM)



ARM Physical IP Enables Comprehensive Power Management!
